

When Homotopy Gets Real

Numerical Decomposition of Real Algebraic Curves and Surfaces in Julia

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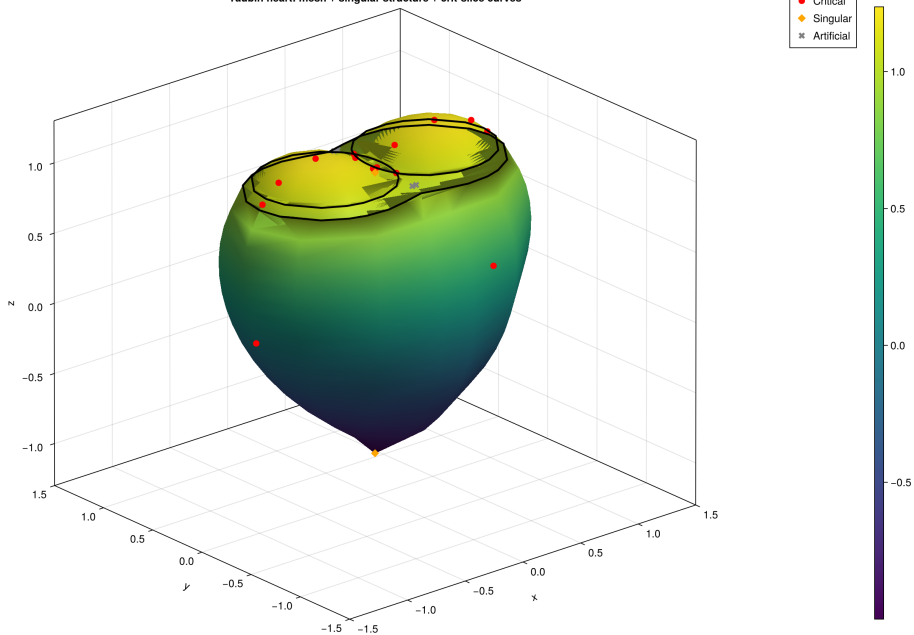
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The problem

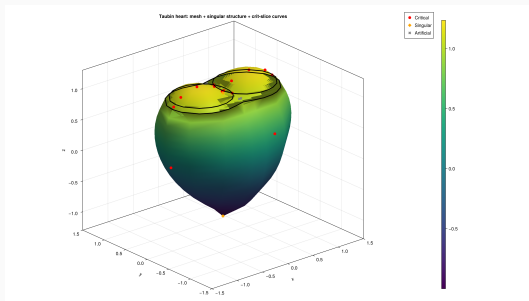
Decompose the real points of an algebraic variety
into a combinatorial, navigable structure:
vertices, edges, faces,
using **homotopy continuation**, not symbolic methods.

Homotopy continuation, in one line: start from an easy system with known solutions, deform it into the real target system, and track the solutions as they deform.

Taubin heart: mesh + singular structure + crit-slice curves



What you just saw



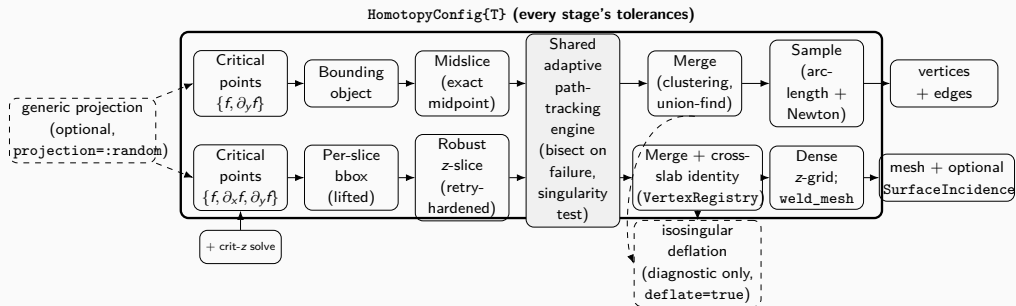
- Surface mesh, vertex decomposition, and critical-slice curves: **one scene**
- Curves, surface, and singular structure computed by the **same** pipeline

Bertini_real (Amethyst et al., 2014, 2017) **HomotopyGetsReal.jl**

- Requires a whole-component witness set
- From Bertini1's **Numerical Irreducible Decomposition** stage
- C++, two-stage pipeline
- Decomposes directly on the **bare** curve/surface equation
- No separate witness-generation stage
- Pure Julia, no C dependency

HomotopyContinuation.jl (Breiding and Timme, 2018) plays the same role here that Bertini1 plays for Bertini_real: the actual path-tracking engine underneath, native Julia instead of C.

Architecture: the six-stage pipeline



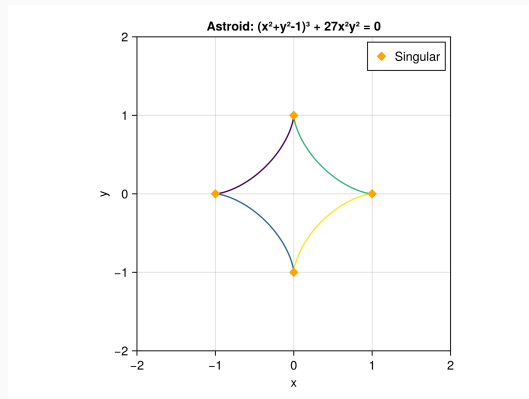
Curve (top) and surface (bottom) lanes share one adaptive path-tracking engine (Step 4, built on `HomotopyContinuation.jl` (Breiding and Timme, 2018)) and one `HomotopyConfig{T}` for every tolerance (outer frame). Both are built on type-stable parametric data structures, `NativeVertex{T}`, `Edge{T}`, and `Face{T}`. Dashed boxes are opt-in.

Curve algorithm: Lu et al. (2007); surface algorithm: Besana et al. (2013) – the same algorithms Bertini_real implements.

Validation: what “correct” means here

- Residuals: how far a computed point sits from the true variety
- Cross-checks against known ground truth: Hauenstein–Wampler examples
- Direct comparison against Bertini_real on shared test cases
- HomotopyContinuation.jl ships its own certifier (certify, Krawczyk/ α -theory-based), not yet used in this pipeline
- 538/538 tests passing, full suite including the slow isosingular-deflation tests

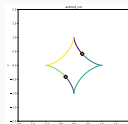
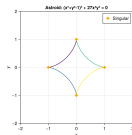
Headline result: the astroid



Same variety, decomposed by both HGR and Bertini_real. Do the counts agree?

Astroid: coordinate match

HomotopyGetsReal.jl Bertini_real



Edge cycle

4

6

Underlying variety

same

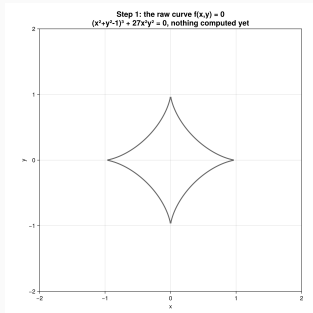
Vertex coordinates

match

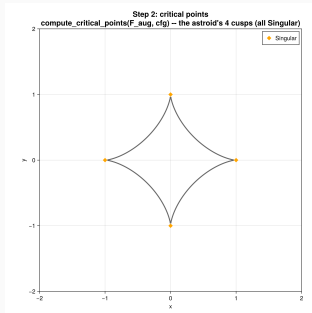
The 6 vs. 4 difference comes from Bertini_real's own random projection: two extra smooth critical points split two of the four arcs. Same underlying curve, finer partition, not extra topology HGR misses (confirmed by exact coordinate matching).

4-cusp, 4-edge topology independently confirmed in Amethyst et al. (2026).

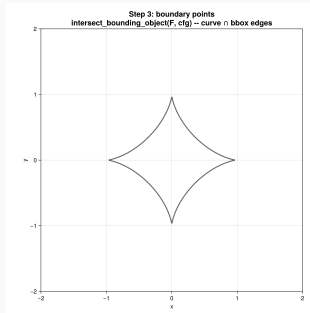
Curve pipeline: raw curve, critical points & boundary points



Step 1: the implicit curve $f=0$, nothing computed yet.



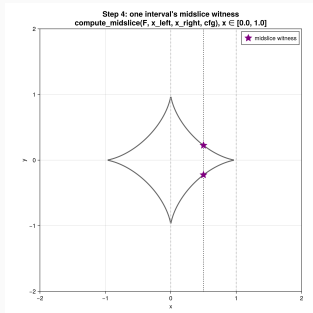
Step 2: 4 vertices found, all Singular, the astroid's 4 cusps.



Step 3: where the curve meets the bounding box, none here; the astroid sits entirely inside it.

```
Faug = System([f, diff(f,y)]; variables=[x,y])  
sols = solve(Faug; show_progress=false)
```

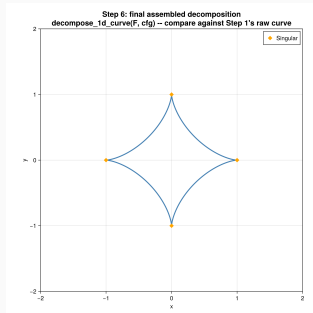
Curve pipeline: midslice witness, connect the dots & final decomposition



Step 4: one interval's midslice witness point, the seed for tracking.



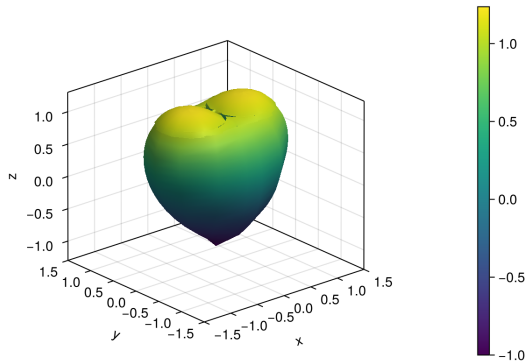
Step 5: tracking bidirectionally from the witness point, over the faded raw curve.



Step 6: full output, 4 vertices, 4 edges. Directly comparable to Step 1.

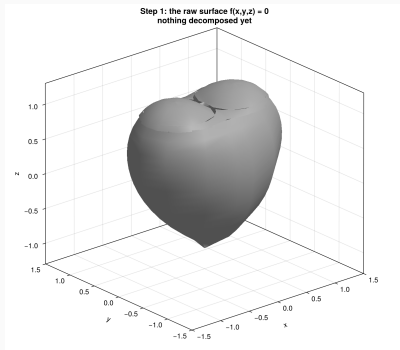
```
result = solve(Fsub; show_progress=false) # witness
ph = ParameterHomotopy(H_sys, [x0], [x0])
track(Tracker(ph), y0, 1.0, 0.0) # connect the dots
```

Case study: the Taubin heart

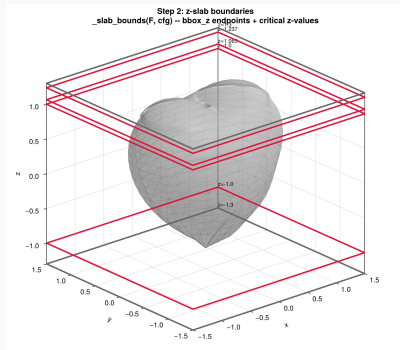


Topologically rich test surface: multiple singularities, non-convex geometry

Surface pipeline: raw surface & z-slab boundaries

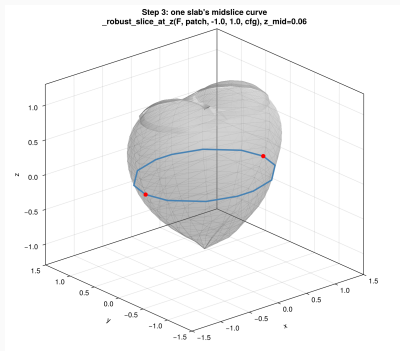


Step 1: the surface, nothing decomposed yet.

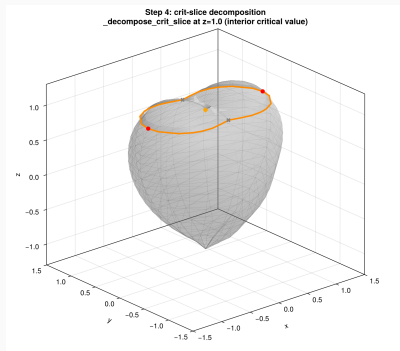


Step 2: z-slab boundaries, bounding-box endpoints plus interior critical z-values.

Surface pipeline: midslice curve & interior critical slice

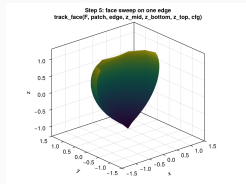


Step 3: one slab's midslice curve, perturbed away from a degenerate naive midpoint.

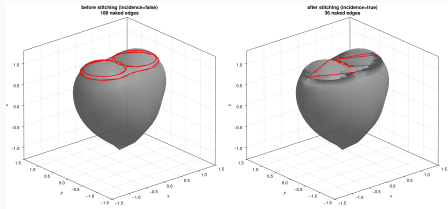


Step 4: the interior case, a critical slice at $z = 1$, not a slab midpoint.

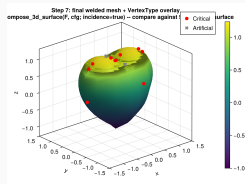
Surface pipeline: face sweep, stitching & final mesh



Step 5: one edge, swept bidirectionally across z into a single face.



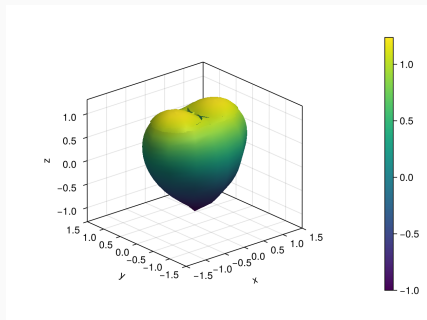
Step 6: incidence stitching, naked edges drop from 188 to the low-to-mid 30s (34 typical).



Step 7: the final welded mesh, the walkthrough's natural endpoint.

```
H_sys = System([f, a*(x-x0)+b*(y-y0)];  
variables=[x,y], parameters=[z])  
ph, tracker = build_tracker(H_sys, z0, cfg; compile=:none)
```

The plain call

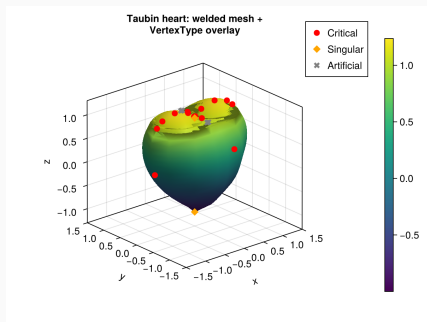


```
vertices, edges, faces, mesh =  
decompose_3d_surface(F, cfg)
```

14 vertices 0 Singular

The default call doesn't surface them. This is default-API behavior, not a correctness failure.

With the incidence overlay

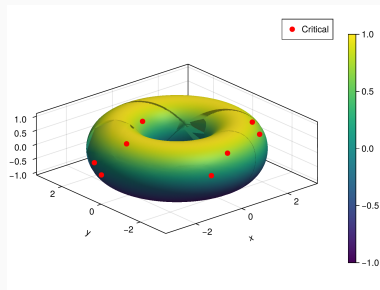


```
verts, edges, faces, mesh, inc =  
decompose_3d_surface(F, cfg;  
incidence=true)
```

20 vertices 2 real Singular cusps

Turning on the incidence overlay recovers the singular structure the plain call hides: cusps localized directly on the mesh, not just reported as coordinates.

The other extension: generic projection



- Torus is **genus-one**, topologically distinct from the other three surfaces (sphere, ellipsoid, Taubin heart, all genus-zero)
- The only one of the four validation surfaces for which the fixed-coordinate default is not merely imprecise but **fails completely**
- Fixed via `projection=:random`, the pipeline's opt-in generic-projection stage

Validation: the residual numbers

Example	n	mean	median	p_{90}	p_{99}	max
Sphere	19,504	2.85×10^{-8}	2.62×10^{-8}	5.32×10^{-8}	7.05×10^{-8}	7.58×10^{-7}
Ellipsoid	19,500	2.70×10^{-7}	2.79×10^{-8}	5.85×10^{-8}	8.99×10^{-8}	1.12×10^{-4}
Taubin heart	1,638	1.034×10^{-7}	3.44×10^{-8}	1.847×10^{-7}	2.000×10^{-6}	2.42×10^{-6}
Torus*	78,340	2.211×10^{-6}	$\sim 1.4 \times 10^{-6}$	5.491×10^{-6}	1.008×10^{-5}	$\sim 2.0 \times 10^{-5}$

Pointwise residuals $|F(p)|$ on production-density meshes (n = mesh points): $|F(p)|$ evaluated at every vertex p of the final welded mesh; p_{90}/p_{99} are the 90th/99th percentile of that distribution. *Torus stats from a partial live run.

Future work

- Isosingular deflation today is **diagnostic only**; it does not yet retain the deflated system for a first-class singular-curve decomposition, the way Bertini_real does
- Blueprint (not implemented): deflate → keep the deflated system → decompose it embedded, seeded by singular-locus samples the surface pipeline already produces
- Bounding uses an **axis-aligned box**: Bertini_real bounds with a sphere instead and scales better doing it, their own manual conceding the box “got messy” at higher dimensions. No architectural reason blocks switching
- Surfaces are hard-capped at **exactly 3 variables** (compute_critical_points throws otherwise): a real gap in demonstrated scale against Bertini_real’s own 14-variable, degree-630 curve, not just this implementation’s current stage

Takeaways

- Every claim in this talk was checked against residuals, published ground truth, or a direct Bertini_real comparison, not asserted
- A substitute for Bertini_real today on single equidimensional hypersurfaces, a complement elsewhere; what's next for this project: a certified critical-point solve
- **A native Julia reimplementation** of Bertini_real's decomposition pipeline: no witness-set stage, with two validated extensions, generic projection and isosingular deflation matched exactly to Hauenstein–Wampler

Amethyst et al. (2014, 2017). Bertini_real. ICMS 2014 / TOMS 2017.

Hauenstein and Wampler (2013). Isosingular Sets and Deflation. FoCM 2013.

Lu et al. (2007). Finding All Real Points of a Complex Curve. 2007.

Besana et al. (2013). Cell Decomposition of Almost Smooth Real Algebraic Surfaces. 2013.

Breiding and Timme (2018). HomotopyContinuation.jl. ICMS 2018.

Amethyst et al. (2026). Kinematic Synthesis over Curves. IMA 2026.

- **HomotopyGetsReal.jl**, v0.2.1, released under the **MIT license**
- Registered: `Pkg.add("HomotopyGetsReal")` works today
- Open source, tested, documented

Thank you

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Implementation assisted by AI coding tools (Cursor, Claude);
all results independently verified.

Questions?

Bonus: Isosingular deflation, the Whitney umbrella

- The benchmark singular point of Hauenstein and Wampler (2013): tip $(0, 0, 0)$ and handle $(0, 0, 1)$
- **Tip:** corank sequence $[3, 2, 0]$, matches the published sequence exactly, terminates at corank 0
- **Handle:** corank sequence $[3, 1, 1]$, matches the published sequence for as many rounds as is computationally feasible; round 5 needs $\sim 5 \times 10^9$ symbolic minors and doesn't terminate
- Deflation is diagnostic today; see Future work, covered earlier, for the path to a first-class singular decomposition